Canny Edge Detection Filter

Concept, Development, Use and Application

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Outline: 1 Edges and the coastline problem · 2 Classical filters · 3 How Canny works · 4 NDWI · 5 Data and method · 6 Results · 7 GeoEdge speed-up · 8 Conclusions

Edges and the coastline problem

What is an edge?

- A boundary where pixel brightness changes rapidly over a short distance, e.g. dark sea water meeting bright dry land
- Ideal shapes: step edge, ramp edge, roof edge (a line edge is a roof edge about 1 pixel wide)
- In satellite images edges are usually ramps over 2–3 pixels: one 10 m pixel can hold both shallow water and wet sand

Why coastlines are hard

- Water boundaries (coasts, riverbanks, lake edges) are used for erosion monitoring, navigation maps and flood levels
- Roads, buildings, concrete docks and field boundaries also have sharp changes, and a simple detector marks them all
- Waves, boat wakes and cloud shadows create false edges inside the water
- Murky water has very weak contrast with the land

Objectives of this paper

1. Explain how Canny works · 2. Compare it with Roberts, Sobel, Prewitt, Laplacian and LoG · 3. Test it on Sentinel-2 water boundaries (SWED) and make it fast on large data

Classical filters and their limits

Convolution: slide a 3 × 3 kernel over the image. Kernel weights sum to zero, so a flat area (open water) gives 0 and an edge gives a large value. Sobel Gx = [−1 0 1; −2 0 2; −1 0 1]. Edge strength $G = \sqrt{(Gx^2 + Gy^2)}$; direction $\theta = \tan^{-1}(Gy/Gx)$.

Filter	Year	Main idea	Main problem
Roberts	1963	Tiny 2 × 2 diagonal kernels, very fast	Easily disturbed by noise and speckle
Sobel	1968	3 × 3, double weight on centre row	Thick edges, several pixels wide
Prewitt	1970	3 × 3, equal weights (averages)	Noise; wide, blurry edges
Laplacian	–	Second derivative, finds zero crossings	Amplifies noise, marks every texture
LoG	1980	Gaussian blur, then Laplacian	Less noise, but rounds corners, misses faint lines

Common problem: they only highlight gradient bands, so there is no clean one-pixel line, and edges break into pieces where they get faint.

Canny’s approach (1986)

Three goals: good detection (real edges, not noise) · good localization (edge at the true centre) · single response (each edge marked once)

1. **Gaussian blur (σ):** removes small speckles that would cause false edges
1. **Sobel gradient:** $G = \sqrt{(G_x^2 + G_y^2)}$ and $\theta = \tan^{-1}(G_y/G_x)$; a ramp edge gives a band several pixels wide
1. **Non-maximum suppression:** keep a pixel only if it is strictly greater than its two neighbours across the edge, so bands become 1-pixel lines
1. **Double thresholding:** above the high threshold = strong, below the low threshold (usually 0.4 × high) = discarded, in between = weak candidates
1. **Hysteresis:** a weak pixel is kept only if it is 8-connected to a strong edge; isolated weak lines are dropped as noise

Why a Gaussian: the best step-edge filter has almost the shape of a Gaussian’s first derivative (about 20% below the exact optimum but much easier in 2D). A wider filter is more reliable in noise but places edges less exactly, so the user must choose σ. Canny is built into OpenCV and MATLAB.

Making Canny find water: NDWI

The problem

- Canny only sees brightness changes and does not know what water is
- On a grayscale satellite image it also gives edges on roads, rooftops, parking lots and farm borders

The fix: use domain knowledge

- Water absorbs near-infrared (NIR) light almost completely; land and vegetation reflect it strongly
- Both reflect visible Green light

NDWI (McFeeters, 1996)

NDWI = (Green − NIR) / (Green + NIR) = (B3 − B8) / (B3 + B8)

Surface	Green vs NIR	NDWI
Water	Green higher	Positive (bright)
Land	NIR much higher	Negative (dark)

- Computing NDWI before Canny erases buildings, roads and crops; the water–land boundary is the main edge left
- MNDWI (Xu, 2006) uses SWIR band B11 (20 m pixels): blocky shorelines, so 10 m NDWI performed better

Data and evaluation method

Data: SWED

- Sentinel-2 Water Edges Dataset, UK Hydrographic Office (Seale et al., 2022); Sentinel-2 images are free, 13 bands, 10/20/60 m pixels
- 256 × 256 tiles, 12 bands on a 10 m grid; hand-drawn masks (water = 1, land = 0) for the test set of 98 images
- Part A: Ashdod port, 15 Feb 2020; 48,394 water and 17,142 land pixels; true coastline 1,730 pixels
- Part B: all 98 scenes (sandy beaches, turbid river mouths, rocky coasts, reservoirs)
- Inputs: Grayscale (B4, B3, B2), NDWI (B3, B8), MNDWI (B3, B11)

Evaluation

- Six filters in MATLAB; each tuned to its best F1 (Canny: 5 σ values × 14 high thresholds = 70 runs, low = 0.4 × high)
- Tolerance 2 pixels (20 m): correct edge within 2 px of the true coastline, extra edge beyond it, missed coastline with no edge nearby
- Precision = correct / (correct + extra); recall = coastline found / whole coastline
- F1 = 2 × precision × recall / (precision + recall); high only when both are high
- Example: Canny marked 1,018 edge pixels, 952 correct, 66 extra, found 1,571 of 1,730 coastline pixels: P 0.94, R 0.91, F1 0.92

Results: six filters on Ashdod (Grayscale)

Filter	Best setting	P	R	F1
Roberts	threshold 0.199	0.78	0.83	0.80
Sobel	threshold 0.215	0.83	0.80	0.81
Prewitt	threshold 0.211	0.83	0.80	0.82
Laplacian	threshold 0.133	0.28	0.98	0.43
LoG	σ = 3; 0.0054	0.79	0.79	0.79
Canny	σ = 3; high 0.50	0.94	0.91	0.92

P = precision, R = recall. 2-pixel tolerance, best setting of each filter.

- **Canny** is best in both precision and recall; thinning gives 1-pixel lines and hysteresis keeps the faint shore connected
- **Roberts, Sobel, Prewitt:** gaps along the docks, small extras on roofs
- **Laplacian:** finds 98% of the coast but only 28% of its edges are coastline
- **LoG:** less clutter, but misses long dock edges

Effect of blur on Canny: σ = 1: P 0.83, R 0.95 (buildings and ships marked) · σ = 3: P 0.94, R 0.91 (best) · σ = 4: R 0.78 (two sides of the narrow breakwater merge). F1 stays high for thresholds of about 0.30–0.60.

Results: effect of input, Ashdod and 98 scenes

Ashdod image (F1)

Filter	Gray	NDWI	MNDWI
Roberts	0.80	0.82	0.78
Sobel	0.81	0.85	0.74
Prewitt	0.82	0.85	0.73
Laplacian	0.43	0.28	0.56
LoG	0.79	0.76	0.62
Canny	0.92	0.89	0.67

98 SWED scenes (mean F1)

Filter	Gray	NDWI	MNDWI
Roberts	0.418	0.732	0.717
Sobel	0.418	0.754	0.673
Prewitt	0.417	0.756	0.676
Laplacian	0.274	0.268	0.374
LoG	0.411	0.724	0.656
Canny	0.486	0.809	0.733

- **Ashdod:** Canny best on Grayscale (0.92) and NDWI (0.89); on MNDWI, Roberts (0.78) beat Canny (0.67). The sea is clear and dark and the port bright, so visible-band contrast is already strong
- **98 scenes:** Canny is best in every column; NDWI + Canny scores highest (0.809). Clean sandy shore (Tile T58LGJ): F1 0.94. Turbid river mouth (Tile T27WVM): 0.82
- **NDWI** lifts Roberts, Sobel and Prewitt from about 0.42 to 0.73–0.76; the 20 m MNDWI band makes the shore blocky; the Laplacian stays poor (0.27–0.37)

GeoEdge: Canny on large images

Problem: a full Sentinel-2 scene is about 10,980 × 10,980 pixels in several bands (over 1.2 GB). Loading it into Python at once uses a lot of memory and can crash a normal computer.

1. **Tiling with an overlap halo:** read 1024 × 1024 blocks, pad each with a 32-pixel margin, run Canny, discard the margin, stitch the centres. No chopped edges at tile borders and no visible seams
1. **C++20 core:** compiler auto-vectorisation (SIMD) and OpenMP, connected to Python with pybind11; processes several pixels per clock cycle
1. **Otsu's method:** picks the high threshold from the gradient histogram (minimises variance within classes); low = 0.4 × high, so no manual guessing

Metric	Baseline	GeoEdge	Gain
Time	47.10 s	11.20 s	4.2×
Speed	2.12 Mpx/s	8.93 Mpx/s	4.2×
Peak RAM	1,840 MB	105.5 MB	17×

Benchmark: 100 million pixels (10,000 × 10,000). Baseline = standard Python/OpenCV.

Conclusions

What we found

- Canny gave the best F1 of the six filters: 0.92 on Ashdod (Grayscale) and a mean of 0.809 on 98 scenes (NDWI)
- On Ashdod: precision 0.94, recall 0.91; across scenes F1 0.94 on a sandy shore and 0.82 in turbid water
- NDWI input lifts the simple filters from about 0.42 to 0.73–0.76
- Best settings depend on the input: $\sigma = 3$ for Grayscale, $\sigma = 1$ for NDWI and MNDWI
- GeoEdge: 4.2× faster and 17× less memory (105.5 MB for 100 million pixels)

Takeaways and limits

- Canny finds all brightness edges, not only coastlines; NDWI supplies the water knowledge and an answer key is needed to score it
- Blur trades reliable detection against exact placement; narrow features such as the breakwater need a small σ
- Remaining errors were inspected in QGIS: intertidal sandbanks, breaking surf and vessel wakes
- A transparent, lightweight alternative to deep-learning segmentation for operational coastal monitoring